Course Wrap-Up

Lecture 5

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Section 1: The “Big Picture”

Step 1 – The "Dev" Loop

Our Workflow

1. Collaboration

- It all starts with a developer and a task.
  - Developer creates a **feature branch**.
  - Developer writes code and **commits**.
  - Developer **pushes** the branch to GitHub.
  - Developer opens a **Pull Request (PR)**.

➢ Key Tool: Git

Step 2 – The "CI" Loop

Our Workflow

2. Continuous Integration

- The Pull Request automatically triggers our CI pipeline
  - GitHub Actions starts a **runner**.
  - Runs npm install.
  - Runs npm test.
  - Builds the **Docker Image**.
  - Pushes the image to Docker Hub.

- Key Tools: GitHub Actions, Dockerfile

Step 3 – The "CD" Loop

Our Workflow

3. Continuous Deployment

❖ After the PR is approved and **merged to main**:

- The CI pipeline runs again on main
- The deploy job is **triggered**
- GitHub Action **SSHes** into our server
- It runs docker-compose pull and docker-compose up -d
- The server is now running the new code

➢ Key Tools: GitHub Actions, Docker Compose

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Section 2: Key Concepts Review

Key Concepts

Part 1

LO1: DevOps Culture

- **CALMS:** Culture, Automation, Lean, Measurement, Sharing.
- Breaking down silos between Dev, Ops, and QA.
- It's a mindset, not just a tool.

LO2: Git Workflow

- **Feature Branch:** `main` is sacred.
- **Pull Request (PR):** The "gate" for code review and CI checks.
- **Merge:** The act of approving and merging code into `main`.

Key Concepts

Part 2

LO3: Dockerfile

- The "blueprint" for your app.
- `FROM`, `WORKDIR`, `COPY`, `RUN`, `CMD`.
- Creates a portable, layered **Image**.
- A **Container** is a running instance of an Image.

LO4: Docker Compose

- A `docker-compose.yml` file.
- Defines and runs **multi-service** apps (e.g., `app` + `db`).
- Creates a shared network.
- Manages volumes for persistent data.

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Key Concepts
Part 3

LO5: GitHub Actions (CI)

- The "automation engine."
- `.github/workflows/ci.yml`
- **Triggers:** `on: [push, pull_request]`
- **Jobs:** `build-and-test`
- **Steps:** `uses: ...`, `run: ...`

LO6: GitHub Actions (CD)

- A second job: `deploy:`
- `needs: build-and-test`
- `if: github.ref == 'refs/heads/main'`
- Uses **Secrets** (`${{ secrets.HOST }}`).
- Runs **SSH** to update the server.

Key Concepts Summary

- **Git**: feature branches + PR reviews
- **CI**: GitHub Actions (npm ci, npm test, Docker build/push)
- **CD**: SSH + docker-compose up -d (main only)
- **Docker Compose**: app + db services
- **K8s**: Pod → Deployment (3 replicas) → Service
- **CALMS**: Culture(Automation+Lean+Metrics+Sharing)